

Retail Data Specialist Portfolio Project

Target Role: Data Specialist - SQL & Excel (Melbourne Retail Analytics Team)

Project Focus: E-Commerce Retail Integrity, Stock Anomalies & Client Query Automation

1. Executive Summary & Strategy

To win this role, your portfolio must prove you can bridge the gap between heavy SQL backends and practical Excel business tools. This project models a high-velocity global retail environment. It showcases your ability to track inventory drift, detect customer data anomalies, and rapidly build analytical dashboards that address cross-functional client queries.

2. Relational Database Schema & Data Generation

Set up a local relational database (PostgreSQL, MySQL, or SQL Server) representing an omni-channel retail setup. Use the following structures:

```
-- Table 1: Customers
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    country VARCHAR(50),
    signup_date DATE
);

-- Table 2: Products
CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    unit_price DECIMAL(10,2),
    warehouse_stock INT
);

-- Table 3: Orders (Simulating Live Transactions)
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date TIMESTAMP,
    status VARCHAR(20), -- 'Completed', 'Pending', 'Cancelled'
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

```
-- Table 4: Order_Items
CREATE TABLE Order_Items (
  item_id INT PRIMARY KEY,
  order_id INT,
  product_id INT,
  quantity INT,
  line_total DECIMAL(10,2),
  FOREIGN KEY (order_id) REFERENCES Orders(order_id),
  FOREIGN KEY (product_id) REFERENCES Products(product_id)
);
```

3. Advanced SQL Queries for Client Query Resolution

The job description highlights 'validating and interpreting client queries'. Implement these two production-grade scripts to prove your technical depth:

Scenario A: Identifying Anomalous Cross-Border Orders (Data Accuracy & Auditing)

```
SELECT
  c.customer_id,
  c.email,
  COUNT(o.order_id) AS total_orders,
  SUM(oi.line_total) AS lifetime_value,
  AVG(oi.line_total) AS avg_order_value
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
JOIN Order_Items oi ON o.order_id = oi.order_id
WHERE o.order_date >= '2026-01-01'
  AND o.status = 'Completed'
GROUP BY c.customer_id, c.email
HAVING SUM(oi.line_total) > 5000
ORDER BY lifetime_value DESC;
```

Scenario B: Inventory Burn Rate & Stock-Out Warning System

```

WITH ProductSales AS (
    SELECT
        product_id,
        SUM(quantity) AS units_sold_30_days,
        SUM(quantity) / 30.0 AS daily_burn_rate
    FROM Order_Items oi
    JOIN Orders o ON oi.order_id = o.order_id
    WHERE o.order_date >= NOW() - INTERVAL '30 days'
    GROUP BY product_id
)
SELECT
    p.product_id,
    p.product_name,
    p.category,
    p.warehouse_stock,
    COALESCE(s.daily_burn_rate, 0) AS daily_burn_rate,
    CASE
        WHEN COALESCE(s.daily_burn_rate, 0) = 0 THEN 'Overstocked / Stagnant'
        WHEN p.warehouse_stock / s.daily_burn_rate <= 7 THEN 'CRITICAL - Stockout under 7 d
ays'
        WHEN p.warehouse_stock / s.daily_burn_rate <= 14 THEN 'WARNING - Restock required'
        ELSE 'Healthy Allocation'
    END AS stock_health_status
FROM Products p
LEFT JOIN ProductSales s ON p.product_id = s.product_id
ORDER BY p.warehouse_stock ASC;

```

4. Excel Interoperability & 'Overlay Analysis'

Export the results of your inventory health query into Microsoft Excel to build the interactive operational asset required by the team.

Step 1: Dynamic Data Structuring & Formula Layout

- * Format your SQL extraction as an official Excel Table named `Tbl_InventoryHealth`.
- * Inject a dynamic pricing tier matrix using an nested XLOOKUP/INDEX-MATCH system to layer regional margins over cost data.
- * Formula Reference: `=INDEX(Ref_Tiers[Margin], MATCH(TRUE, Ref_Tiers[MinStock] <= [@warehouse_stock], 0))`

Step 2: Pivot Table Configuration & Executive Views

- * Build a dedicated analytics tab containing a Pivot Table categorized by *Product Category* and *Stock Health Status*.

- * Display dual metrics: Total Capital Tied Up (Sum of Product Cost * Stock) and Total Count of Critical Stockouts.
- * Connect an interactive Slicer for *Category* to simulate rapid ad-hoc troubleshooting during team synchronization meetings.

5. Tactical Action Plan & Time Horizons

Phase / Horizon	Core Focus Area & Technical Actions
Days 1–2: Database Setup	Provision local SQL environment. Populate tables with at least 5,000 synthetic retail order sequences to test relational scale.
Days 3–4: Analytics Writing	Execute and validate data sanitization queries. Isolate empty rows, date mismatches, and flag negative inventory discrepancies.
Days 5–6: Excel Synthesis	Pipe data to Excel. Configure conditional rules (Red fills for critical items), cross-sheet lookup logic, and executive Pivot views.
Day 7: Deployment Prep	Commit clean, heavily documented scripts to GitHub. Record a 3-minute video overview explicitly mapping project pipelines to the role requirements.